C–0037

Sub. Code
11813

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023.

First Semester

Aeronautical Science

MATHEMATICS – I

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Find out all the Eigen values of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

2. If -1 and 3 are Eigen values of $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ then find the Eigen values of $-3A^{-1}$.

3. Show that the sphere whose centre is $(1, 2, -2)$ and radius 3 passes through the origin.

4. Find the angle between the radius of curvature in terms of parametric co-ordinate systems.

5. Write down the formula for radius of curvature in terms of parametric co-ordinate systems.

6. Find the envelope of the curve $y = mx + \frac{a}{m}$, where m is a parameter.

7. Write the tailor's series of $f(x, y)$ at the point (a, b).

8. If $u = x^2$ and $v = y^2$ find $\frac{\partial(u,v)}{\partial(x,y)}$.

9. Transform the equation $(x^2D^2 - xD + 4)y = x^2 \sin(\log x)$ into linear equation with constant coefficient.

10. Find the P.I for $(D^2 + 2D + 1)y = \cos 2x$.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Verify Cayley Hamilton theorem for the matrix $\begin{bmatrix} 1 & 0 & -2 \\ 2 & 2 & 4 \\ 0 & 0 & 2 \end{bmatrix}$.

Or

(b) Find the Eigen values and Eigen vectors for the matrix $A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$.

12. (a) Prove that the points $(2,-2,-1)$, $(2,3,4)$, $(-3,5,1)$, $(4,-1,2)$ are co-planar.

Or

(b) Find the equation of the plane which passes through the points $(6,2,-4)$ and $(3,-4,1)$ and is parallel to the line joining the points $(1,0,3)$ and $(-1,2,4)$.

13. (a) Find the radius of curvature of the curve $r = a(1 + \cos\theta)$ at the point $\theta = \frac{\pi}{2}$.

Or

(b) Find the equation of the circle of curvature of the parabola $y^2 = 12x$ at the point $(3, 6)$.

14. (a) If $u = f(x, y)$ where $x = r\cos\theta$ and $y = r\sin\theta$ then prove that $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$.

Or

(b) Find $\frac{du}{dt}$ if $u = x^3y^2 + x^2y^3$, where $x = at^2$; $y = 2at$.

15. (a) Solve $(D^4 - 2D^3 + D^2)y = x^2 + e^x$.

Or

(b) Solve $(x^2D^2 - 2xD - 4)y = 32(\log x)^2$.

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Diagonalise the matrix $A = \begin{bmatrix} 2 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix}$ by using orthogonal transformation.

Or

(b) Find the centre and radius of the circle given by $x^2 + y^2 + z^2 - 2x - 4y - 6z - 2 = 0$ and $x + 2y + 2z - 20 = 0$.

17. (a) Find the evolute of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Or

(b) Discuss the maxima and minima of the function $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

18. (a) Solve the simultaneous equations $\frac{dx}{dt} + 2x - 3y = 5t$; $\frac{dy}{dt} - 3x + 2y = 2e^{2t}$ (Find any one X or Y).

Or

(b) Reduce the quadratic form $2x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 - 2x_1x_3 - 4x_2x_3$ to canonical form by orthogonal transformation.

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C–0039

Sub. Code
11823

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023.

Second Semester

Aeronautical Science

MATHEMATICS – II

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Evaluate $\int_1^b \int_1^a \frac{dxdy}{xy}$.

2. Find the area between the Circle $x^2 + y^2 = a^2$ and the $x + y = a$ lying in the first equation by double integration.

3. Prove that is irrotational.

4. If are irrotational then prove that is solenoidal.

5. State any two properties of analytic function.

6. Verify the function is harmonic or not.

7. Find the Laplace transform of $\left[\frac{1}{\sqrt{t}}\right]$.

8. Verify initial value theorem for the function.

9. Show that the correlation coefficient lies between –1 and 1.

10. The following data represent the number of accidents in a place for the past week. Find its standard deviation. The data's are 7, 9, 12, 15, 5, 4, 11.

Part B $(5 \times 5 = 25)$

Answer **all** questions

11. (a) Evaluate $\int_0^1 \int_0^1 \int_0^2 xyz \, dzdydx$

Or

(b) Why do we change the order of integration in multiple integrats? Justify your answer with an example.

12. (a) Verify that is irrotational further find also its corresponding scalar potential.

Or

(b) Verify stokes's theorem for the vector field where S is the surface of upper hemisphere and C is its boundary in xy plane.

13. (a) If $f(z)$ is analytic, show that $f(z)$ is constant if real part of $f(z)$ is constant.

Or

(b) Find the bilinear transformation which maps the points into the points.

14. (a) Find Laplace transforms of the function $\sin\sqrt{\frac{t}{I}}$.

Or

(b) Find the Laplace transform of $\sqrt{\frac{t}{\pi}}$ and hence find $L\left\{\frac{I}{\sqrt{\pi t}}\right\}$.

15. (a) Calculate the Spearman's Rank correlation from the following data.

Marks in Tamil	75	40	52	65	60
Marks in English	25	42	35	29	33

Or

(b) Calculate the two regression lines X on Y and Y on X from the following data.

X	12	14	15	14	18	17
Y	42	40	45	47	39	45

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Transform the integral into polar co-ordinates and hence evaluate.

Or

(b) Verify Gauss Divergence theorem for the Vector function over the cubic region bounded by the line.

17. (a) Prove that is harmonic and hence find the analytic function.

Or

(b) Find the image of the circle $|Z-1|=1$ in the complex plane under the mapping $W=\frac{1}{z}$.

18. (a) Find the correlation coefficient from the data

Type A	12	9	8	10	11	13	7
Type B	14	8	6	9	11	12	3

Or

(b) Solve the difference equation given that by Laplace transform.

C–0040

Sub. Code
11824

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Second Semester

Aeronautical Science

ENGINEERING MECHANICS AND STRENGTH OF MATERIALS

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define visual work bodies.

2. State law of parallelogram of forces.

3. Explain D'alembert's principle.

4. Define centroid.

5. Define velocity ratio.

6. Define mechanical efficiency.

7. How to identify whether a frame is perfect or imperfect?

8. What is the classification of frames?

9. Define stress.

10. State Hooke's law.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Define the following

    (i) Moment of forces (3)

    (ii) Define mechanics. (2)

Or

(b) Explain moment of Inertia with Q diagram.

12. (a) What are the equivalent forces acting on moving object?

Or

(b) Explain the principle of virtual work.

13. (a) Explain about effect of rolling resistance.

Or

(b) Distinguish between journal bearing and thrust bearings.

14. (a) What are the assumptions should be made while solving truss problems?

Or

(b) Explain various types of frames.

15. (a) What is the limitation in the Euler's formula?

Or

(b) A close-coiled helical spring is required to carry a load of 150 N. If the mean coil diameter is to be 8 times that of the wire, calculate these diameters. Take maximum shear stress as 100MPa.

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Explain system of fundamental units and derived units.

Or

(b) Explain about newtons law of motions.

17. (a) Explain the Friction on Inclined plane, when force is pulling upwards.

Or

(b) Explain the theory of Friction on wedges.

18. (a) Explain the method of drawing mohr's circle of principal stresses and strains.

Or

(b) A cylindrical vessel 2 m long and 500 mm in diameter with 10 mm thick plates is subjected to an internal pressure of 3 MPa. Calculate the change in volume of the vessel. Take E = 200 GPa and Poisson's ratio = 0.3 for the vessel materials.

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C–0041

Sub. Code
11832

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Third Semester

Aeronautical Science

THERMODYNAMICS

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is indicated power?

2. What is network done by system?

3. What is known as Thermal Efficiency of Heat Engine Cycle?

4. What is called heat pump cycle or refrigeration cycle?

5. What is called polytrophic process?

6. What is called volumetric efficiency?

7. What is calorific value of fuel?

8. What are the combustion instabilities?

9. Define specific impulse?

10. Write the significance of vane type compressor?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What do you understand by shaft work?

Or

(b) Write short notes on Latent Heat.

12. (a) Write short notes on volumetric efficiency.

Or

(b) Explain the production of dry ice flow diagram.

13. (a) What is polytrophic process and its significance?

Or

(b) Write short notes on entropy of Gas mixtures.

14. (a) Write the importance of calorific value.

Or

(b) What are the problems associated with incomplete combustion?

15. (a) Write short notes on Positive displacement compressor.

Or

(b) What are the advantages and disadvantages of closed cycle?

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Derive the enthalpy equation and its significance.

Or

(b) What are the causes of irreversibility?

17. (a) Briefly explain about the Vapor compression refrigeration cycle.

Or

(b) Briefly discuss about the Multistage Compression.

18. (a) Discuss about the rotary screw compressors.

Or

(b) List out and explain the basic components of gas turbine engine.

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C–0042

Sub. Code
11834

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Third Semester

Aeronautical Science

AERODYNAMICS AND HELICOPTER THEORY

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is Bernoulli's principle?

2. Define Angle of Attack.

3. Write the Lift and Drag equations.

4. What is the function of Slats?

5. What is Blade Flapping?

6. What is Coriolis effect?

7. What is Semi rigid rotor?

8. Define Tracking.

9. Write a short note on Swash Plate.

10. What is the purpose of Freewheeling unit?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the layers of atmosphere and its assumptions?

Or

(b) Discuss the factors contributing Lift generation.

12. (a) Explain the different lateral stability of an aircraft.

Or

(b) What are the types of drag experienced by an aircraft? Explain in detail.

13. (a) With the help of a diagram, explain the major parts of a helicopter and their functions.

Or

(b) What do you mean by Auto rotation?

14. (a) Describe the procedure for Blade Alignment.

Or

(b) Explain the flight control system of a helicopter.

15. (a) What is the function of tail rotor drive shaft?

Or

(b) Write short notes on tail rotor gear box system.

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Explain the following terms:

    (i) Temperature

    (ii) Viscosity

    (iii) Pressure

    (iv) Humidity

Or

(b) Explain the effects of various control surfaces with roll, pitch and yaw motion of an aircraft.

17. (a) Explain the terms:

    (i) Translating tendency

    (ii) Translational lift

Or

(b) Explain in detail about rotor head maintenance.

18. (a) Describe the span wise dynamic balance of the main rotor blade.

Or

(b) Explain the pitch changing mechanism of tail rotor.

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C–0043

Sub. Code
11835

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Third Semester

Aeronautical Science

AIRCRAFT CONSTRUCTION

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Explain the purpose of slats.

2. Explain safe life design.

3. What is the need for the control surface in the aircraft?

4. What is the function of spoilers?

5. Name four types of brakes used in aircraft.

6. What is antiskid system?

7. What are the sources where weight and balance data can be obtained from?

8. What is the importance of aircraft balance?

9. Write short note on levelling the aircraft with Plum bob.

10. Define washin and washout.

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Define aircraft station numbers. Write in detail about any two of them.

Or

(b) What is cantilever wing? Describe its construction.

12. (a) Describe the principles and operational details of Fly-by- wire system.

Or

(b) Explain the types of tabs.

13. (a) Explain about the nose wheel steering system.

Or

(b) Explain the following:

(i) Torque link

(ii) Swivel gland

14. (a) List out the weighing procedures of an aircraft.

Or

(b) How will you compute the centre of gravity location for various aircraft?

15. (a) Write short notes on fuselage alignment check.

Or

(b) Write down the procedure of the levelling of the aircraft.

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Differentiate between 'safe life' and 'fail safe' in structural design philosophy. Describe the sandwich construction in detail.

Or

(b) Write in detail about the types of the aircraft fuselage with suitable diagram.

17. (a) Discuss in detail about equipment used to weighing the aircraft.

Or

(b) Explain the electrical retraction system of aircraft in landing gear.

18. (a) Explain the operation of aircraft brake system.

Or

(b) List out how you will calculate Empty Weight Center of Gravity (EWCG) for various aircrafts.

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C–0044

Sub. Code
11842

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Fourth Semester

Aeronautical Science

AIRCRAFT SYSTEMS

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. State Pascal's law.

2. Write the significance of piston type accumulator.

3. Write the role of silica gel in pneumatic system.

4. What are the sources of pneumatic supply?

5. Write the significance of pressure controller.

6. List out the types of heating in air-conditioning.

7. What is wind shield wiper?

8. Write the significance of Rain Repellent system.

9. List out the types of Aviation Fuel.

10. What is the integrated fuel tank and its significance?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain the double spool selector valve.

Or

(b) List out the types of actuators and its applications.

12. (a) What type of compressor used in pneumatic system and explain it?

Or

(b) Write short note on Pressure controller used in pneumatic system.

13. (a) What is the need for pressurisation system and its basic components?

Or

(b) Explain about ducting arrangement of air-conditioning distribution system.

14. (a) Write short note on Pneumatic De-icer boots operation.

Or

(b) List out the properties of Aviation Fuel.

15. (a) Write short notes on:

(i) Fuel Jettisoning

(ii) Fuel heating.

Or

(b) Explain about the any one type of wind shield anti-icing method.

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) List out and explain the various pressure control devices used in hydraulic system.

Or

(b) Advantages and disadvantages of hydraulic system.

17. (a) With neat sketch, explain about the operation of bootstrap cooling system.

Or

(b) Write short note on:

(i) Operation of pressure controller

(ii) Temperature control system in air-conditioning system.

18. (a) With neat sketch, explain about the Gravity Feed Fuel system.

Or

(b) (i) Cross feed valve

(ii) Transfer valve

(iii) Re-fuel/De-fuel valve.

C–0045

Sub. Code
11843

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Fourth Semester

Aeronautical Science

AIRCRAFT INSTRUMENTS

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. State the fundamental properties of gyroscope.

2. What is meant by true altitude and absolute altitude?

3. Define ISA.

4. What is the basic difference between LED and LCD?

5. What are Q codes?

6. What are the different types of aircraft pressure gauge?

7. What are TAS and EAS?

8. How can you measure RPM of engine?

9. What are the navigation instruments?

10. What does the green arc on an aircraft temperature gauge indicates?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain the working principle of displays used in civil and military aircrafts.

Or

(b) State the principle, advantages and disadvantages of HUD, HMD.

12. (a) How does an altimeter work?

Or

(b) What are the common errors associated with altimeters?

13. (a) What are basic difference and common features in the Artificial Horizon and Turn and slip Indicator.

Or

(b) Explain about Gyroscopic Instruments.

14. (a) Explain mach meter and capacitor type fuel level indicator.

Or

(b) Describe the construction of Wheatstone bridge and its type.

15. (a) Explain the laws of magnetism.

Or

(b) Explain the constructional features of RR Compass.

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Explain the needs for avionics system.

Or

(b) Explain with a neat sketch about pitot static system.

17. (a) Explain in detail Airspeed indicators.

Or

(b) What type of indicating system is used to measure the turbine-engine exhaust gas temperature?

18. (a) Explain three types of basic instruments used in an aircraft flight station?

Or

(b) Describe the two types of Angle of Attack sensors used in aircraft.

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C–0046

Sub. Code
11844

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Fourth Semester

Aeronautical Science

AIRCRAFT MATERIALS, HARDWARE AND NDT

(2016 onwards)

Duration: 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Classify carbon steels based on its carbon content.

2. Define Modulus of Elasticity.

3. What is the purpose of heat treatment?

4. What is Anodizing?

5. What is the purpose of blind bolt?

6. What are dopes and doping agents?

7. What are composite materials?

8. What are Fibre glass?

9. Write down the advantages of Boroscope inspection.

10. Write the principle used in Ultrasonic testing.

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe the composition and properties of aluminium alloy 7075-T6.

Or

(b) Explain the terms :

(i) Rigidity

(ii) Brittleness.

(iii) Yield strength.

12. (a) What are the objectives of tempering and discuss briefly the tempering process?

Or

(b) What is Corrosion? How Corrosion can be prevented? Explain any two methods.

13. (a) Discuss about the seasoning of woods.

Or

(b) What is the difference between a standard machine screw and a structural screw?

14. (a) Write down the functions of the Matrix phase in composite materials.

Or

(b) Describe honey comb core materials.

15. (a) Discuss the operational procedures involved in Izod test.

Or

(b) List the methods of Non destructive testing which are commercially available. Describe one method in detail.

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Explain in detail about the properties of copper and its alloys.

Or

(b) Discuss the different annealing process. Is spherosiding different from annealing?

17. (a) Describe briefly the different methods of case hardening of steel.

Or

(b) Explain in detail about the riveting procedure.

18. (a) Explain any one manufacturing process for composite structures.

Or

(b) What is creep in metals? Draw a typical creep curve and explain the three stages of creep.

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C–0047

Sub. Code
11851

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Fifth Semester

Aeronautical Science

AIRCRAFT RULES AND AIRWORTHINESS REGULATIONS

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What are Airworthiness advisory Circulars?

2. Write the types of Log Books used in aircraft.

3. Write the validity of Certificate of Airworthiness.

4. Mention about the Registration Fees of Aircraft.

5. Define Defect.

6. What are the types of weighing machines used for aircraft?

7. Define MMEL.

8. What are the roles of Flight Data Recorder?

9. List out the basic instruments for day flying of aircraft.

10. Mention the various grades of oils used in aircraft.

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the procedures to be followed during the modification of equipment?

Or

(b) How operator will gel the revisions and Amendments of CAR?

12. (a) Mention about few Procedures for renewal of AME License.

Or

(b) List out the occasions for cancellation of C of A.

13. (a) Mention the few procedures for rectification and analysis of defect in aircraft.

Or

(b) Write short notes on Weight and Balance control of aircraft.

14. (a) List out sonic of First Aid kits.

Or

(b) What is significance of TCAS?

15. (a) List out the procedures of storing of oils and Lubricants.

Or

(b) What are precautionary measures to be following while the aircraft is in storms conditions?

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Briefly explain about the procedures for renewal of AME License.

Or

(b) List out the minimum requirements for renewal of Approved Organization of category in Stores.

17. (a) What is minimum equipment list and its approval?

Or

(b) Explain about the Universal Precaution List.

18. (a) List out the procedures for defueling of aircraft.

Or

(b) Briefly explain the privileges of various categories of AME's.

C–0048

Sub. Code
11852

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Fifth Semester

Aeronautical Science

PISTON ENGINE AND PROPELLER

(2016 Onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. List out the names of materials used for Exhaust valve.
2. Define Engine Efficiency.
3. What are the purposes of Piston rings?
4. Define Crankshaft.
5. List out the types of aviation fuels.
6. List out the requirements of lubricating oil.
7. What is Magneto?
8. What is the importance of spark plugs pressure testing?
9. List out the types of propeller controls.
10. What types of wood used for wooden propeller?

Part B

$(5 \times 5 = 25)$

Answer **all** questions

11. (a) Draw and derive the Otto Cycle P-V diagram and its efficiency.

Or

(b) What is role of firing order of Piston Engine?

12. (a) Write the functions of piston engine cylinder and its types.

Or

(b) Write short notes on Piston engine accessory section.

13. (a) Write short notes on Carburetor Icing and its importance.

Or

(b) Write the classification of Lubricating oil.

14. (a) Write short notes Clutch Mechanism of piston engine.

Or

(b) Write short notes on spark plug construction details.

15. (a) What is propeller feathering?

Or

(b) What is propeller efficiency and its importance?

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Briefly discuss about the development of piston engines.

Or

(b) Explain the construction features of propeller shaft.

17. (a) List out the maintenance procedure of float type carburetor.

Or

(b) With neat sketch, explain the Gravity Feed fuel system of Piston Engine.

18. (a) List out the maintenance procedure of Spark Plug Maintenance.

Or

(b) List out the Propeller system malfunctions.

C–0049

Sub. Code
11853

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Fifth Semester

Aeronautical Science

GAS TURBINE ENGINE

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What do you understand by the term air breathing engines?

2. What is a scramjet engine?

3. Define thrust.

4. What are the functions of an exhaust nozzle?

5. What are the two basic classifications of a propeller?

6. Define the blade angle.

7. What is the need of electronic engine control?

8. What is the difference between a supervisory EEC and full authority EEC?

9. What are types of synthetic oils?

10. On what type of aircraft is auto ignition used and how it is activated?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) State and explain the working principle of turbojet engine.

Or

(b) Discuss the factors affecting the thrust and performance of gas turbine engine.

12. (a) Illustrate the working principle of centrifugal compressor with neat sketches.

Or

(b) Briefly explain the supersonic inlets.

13. (a) Describe the forces acting on a propeller in flight.

Or

(b) Describe the function of a toque meter.

14. (a) Discuss the properties and characteristics of jet fuel.

Or

(b) Explain the function and working of FADEC system with neat sketch.

15. (a) Describe the main components of a gas turbine engine ignition system.

Or

(b) Describe the function of an APU

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Briefly explain the methods of thrust augmentation.

Or

(b) What are the advantages and disadvantages of a ramjet engine and what are its applications?

17. (a) Explain the various methods of thrust reversing.

Or

(b) Describe the principle of operation for the dowry R321 turbo propeller.

18. (a) Describe the basic design of a fuel control for a turbo shaft engine.

Or

(b) Describe the basic principle of operation of operation of a total loss oil system.

C–0050

Sub. Code
11854

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Fifth Semester

Aeronautical Science

AIRCRAFT ELECTRICAL SYSTEM

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. How do you determine the direction of current flow?

2. What is a solenoid?

3. What are the active materials in lead-acid storage cell?

4. What are the types of charging equipment?

5. What is commutation?

6. Write some advantages of brushless alternator?

7. What is meant by negative ground system?

8. What is bonding jumper?

9. What does spin up cause in antiskid control system?

10. Write some application of lights in external lighting.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Compare a solenoid with an electromagnet.

Or

(b) Describe the principles of switch installation.

12. (a) Describe the construction of nickel cadmium batteries.

Or

(b) Explain how battery inspection.

13. (a) Explain permanent magnet generators with neat diagram.

Or

(b) Describe about the equalizing circuit.

14. (a) Explain in briefly about parallel- electrical system with diagram.

Or

(b) Explain how electric wires are identified in a system.

15. (a) Explain briefly about the construction of navigation light circuit.

Or

(b) Describe about turbine engine auto ignition circuits.

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Assume that we wish to install a wire in a bundle 30 ft (8.85 m) long to carry a continuous load of 10A in a 28 V circuit?

Or

(b) Describe the principles of voltage regulation.

17. (a) Describe how wire bundles should be routed in an aircraft.

Or

(b) Explain briefly about the circuit breakers with neat sketch.

18. (a) Describe a power distribution system for a large aircraft.

Or

(b) Describe the theory of operation for a transport category aircraft brushless alternator.

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C–0051

Sub. Code
11855

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Fifth Semester

Aeronautical Science

INDUSTRIAL MANAGEMENT

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Write some characteristics of Management in Organization.

2. What are the benefits of MBO?

3. Define Organizing.

4. What is Manpower Planning?

5. Define Motivation.

6. Write the duties of a Supervisor.

7. What are the types of Decisions?

8. How do you manage records?

9. What is the role of Inventory Management?

10. What are the uses of Work Study?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain in detail about Mintzberg Management roles.

Or

(b) Briefly discuss about Strategic Planning and Operational Planning.

12. (a) Briefly discuss the characteristics and functions of Organization.

Or

(b) Differentiate Line and Staff Authority.

13. (a) Explain the process of Decision Making with the help of an example.

Or

(b) Describe Theory X and Theory Y.

14. (a) What do you mean by Time Management? Explain the techniques of time management.

Or

(b) What are the functions of PPC?

15. (a) Explain the various inventory control methods.

Or

(b) Define ergonomics and design relation in work posture.

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) What is planning? Explain the steps involved in planning?

Or

(b) Write about the advantages of manpower forecasting.

17. (a) Elucidate the steps in selection process.

Or

(b) What are the various barriers to communication? How will you overcome those barriers?

18. (a) Explain in detail how to manage office correspondence.

Or

(b) Elaborate on the procedure for carrying out a work study.

C–0052

Sub. Code
11861

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Sixth Semester

Aeronautical Science

AIRCRAFT MAINTENANCE, GROUND HANDLING AND SUPPORT EQUIPMENTS

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is Maintenance?

2. What is the significance of maintenance records?

3. What are the types of landing gear?

4. What is the purpose of the struts in an aircraft?

5. Write down the classification of the structural damage.

6. What is rivet layout?

7. What is difference between taxing and towing?

8. What is meant by Material safety data sheet?

9. What is meant by Heat Exchanger?

10. What is APU?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe different types of publications used in the aircraft maintenance.

Or

(b) Explain various records maintained by operator.

12. (a) Discuss about the landing gear retraction test procedure.

Or

(b) Write short notes on lightning strike.

13. (a) Briefly describe about the sheet metal bend.

Or

(b) Write short notes on safety tools.

14. (a) Explain about tie down procedure for multi engine aircraft and helicopter.

Or

(b) Explain about the taxing signals.

15. (a) Briefly explain Air-conditioning and Heating Unit.

Or

(b) Discuss about the air start unit.

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) What are the types of the inspection of the aircraft? Briefly discuss about them.

Or

(b) Briefly explain the maintenance and inspection of the landing gear.

17. (a) Briefly explain about the longeron, spar, rib repair

Or

(b) Write down the precautions to be followed while towing the aircraft.

18. (a) Explain in detail about aircraft fueling operation and precautions need to be followed.

Or

(b) Explain briefly about the types of rivet failures.

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C–0053

Sub. Code
11862

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Sixth Semester

Aeronautical Science

AERO ENGINE MAINTENANCE

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is Engine Overhaul?

2. What is the purpose of engine compression test?

3. What kinds of damages occur in the propeller blades?

4. How do you stop Propeller cavitations?

5. What are the checks to be carried out in ice-forming conditions?

6. List out the type's ignition units used in piston engine?

7. What is combustion stability?

8. Write the functions of exhaust nozzle?

9. What are the effects of detonation?

10. Write the significance of "Duration of rundown"

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) List out the effects of excessive temperature in aircraft engine?

Or

(b) Write short notes on Piston Rings?

12. (a) List out the materials used for engine propellers?

Or

(b) Write the basic inspections of propeller blade?

13. (a) What is Post stopping of engine?

Or

(b) List out the basic checks required for ignition system of piston engine?

14. (a) Write short notes on "Inspections of turbine blades"?

Or

(b) List out the inspections required in turbine nozzle?

15. (a) What is thermocouple principles?

Or

(b) List out the parameters considered for assessment of engine performance?

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Describe about the Propeller blade Tracking Check.

Or

(b) List out the checks required for inspections of Piston engine cylinder head.

17. (a) Describe the inspections for combustion chamber.

Or

(b) Explain the internal conditions checking of engine

18. (a) Write short notes on

(i) main parts of ignition harness

(ii) spark plug inspection.

Or

(b) List out the checks list for "Foreign Object Damage".

C–0054

Sub. Code
11863

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Sixth Semester

Aeronautical Science

AIRCRAFT COMMUNICATION AND NAVIGATION SYSTEM

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Write down the characteristics of sky wave?

2. Define Amplitude modulation?

3. For what purpose are VHF communication employed?

4. What is the functions of SATCOM?

5. What is the possible range of a VLF navigation station?

6. What are the advantages of MLS?

7. What is the purpose of ground proximity warning system?

8. What is the function of ATC transponder?

9. What is the nature of Radar signal?

10. What is meant by radome?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain about the Amplifiers and its types?

Or

(b) Explain with the block diagram of functions of radio receiver?

12. (a) Briefly explain the audio control system?

Or

(b) Write the procedure to testing of a communication radio?

13. (a) Write the theory of operation of Automatic Direction Finder (ADF)?

Or

(b) Write the theory of operation of Distance Measuring Equipment?

14. (a) Describe the Emergency Locator Transmitter (ELIT)?

Or

(b) Explain about the operation of TCAS?

15. (a) Describe the principle unit of analog radar system?

Or

(b) Explain about Doppler radar?

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Explain the working operation of RMI with neat sketch?

Or

(b) Describe the radar safety.

17. (a) Explain how do installation of radar system?

Or

(b) Explain the working operation and principle of ILS?

18. (a) Describe the clear function of the weather mapping system display with example?

Or

(b) Explain the various types of modulation?

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C–0055

Sub. Code
11864

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023.

Sixth Semester

Aeronautical Science

AIRPORT AND AIR TRAFFIC SERVICES

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What are the objectives of ICAO?

2. What are the functions of Airport?

3. Differentiate stop way and clear way.

4. What is load classification number?

5. Define Holding Apron.

6. What is the difference between terminal and gate?

7. List out the airport markings.

8. What is 'X-ray' unit? State its purpose.

9. What is meant by controlled Airspace?

10. What are the objectives of ATS?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain about different categories of Airport.

Or

(b) Explain about Air freedom rights.

12. (a) Write a short note on Primary and Secondary runway.

Or

(b) Explain about Taxiway marking with neat diagram.

13. (a) Describe the terminal passenger flow in detail.

Or

(b) Write a short note on Airport Congestions and Delay.

14. (a) Describe the Taxiway lighting system in detail.

Or

(b) Explain in detail about Lounge area requirements.

15. (a) Briefly explain flight plan and its content.

Or

(b) Describe the operation of marker beacon and ILS.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Explain ICAO and its functions.

Or

(b) Explain the environmental factors to be considered with respect to Airport operation.

17. (a) Narrate various sources of Airport Revenue.

Or

(b) Describe in detail various systems employed in airport security.

18. (a) Explain VASI and PAPI with neat sketch.

Or

(b) Describe the establishment, designation and identification of units providing ATS.

C–0056

Sub. Code
11865

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023

Sixth Semester

Aeronautical Science

TRAVEL AND TOUR MANAGEMENT

(2016 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. List out the types of tour operators?

2. Write the concept of Free Independent traveler?

3. Define travel trade.

4. What do you understand the travel agency sector?

5. List out the components of attraction of tourism?

6. Write the role of Travel Documentation Department?

7. List out the minimum requirements of travel agency?

8. What is travel protection?

9. Write the basic classification of hotels?

10. Write the role of ITDC?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write short notes on Technology Driven period of Thomas Cook.

Or

(b) List out the areas of relation to tour operators.

12. (a) Write short notes on foreign Exchange.

Or

(b) What do you understand from International ticking department?

13. (a) List out and explain the basic components of tourism.

Or

(b) Write short notes on types of tour packages.

14. (a) Write the concept of Franchise.

Or

(b) What are the roles of IATO?

15. (a) Write the organization structure of ITDC.

Or

(b) What are the opportunities provided by the TAAI?

Part C

$(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Briefly describe about the integration and linkages in the travel operation.

Or

(b) Explain the different functions in travel operation.

17. (a) Explain the importance of itinerary development in travel operation.

Or

(b) List out the types of accommodation.

18. (a) List out the factors affecting the tour package formulation.

Or

(b) Explain the roles and functions of IAAI.

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C–1214

Sub. Code
11813

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023.

First Semester

Aeronautical Science

WORKSHOP PRACTICES

(2023 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 1 = 10)$

Answer **all** questions.

1. Why is it important to have good lighting in a workshop?
   (a) To save on electricity costs
   (b) To create a cozy atmosphere
   (c) To ensure safe and efficient work
   (d) To minimize distractions

2. Which of the following is NOT a good practice for arranging tools in a workshop?
   (a) Storing tools in a way that is easy to access and use
   (b) Keeping sharp tools in a separate area from other tools
   (c) Leaving tools on the floor or work surfaces
   (d) Putting tools back in their designated places after use

3. Which of the following electrical test instruments is used to measure voltage?
   (a) Multimeter (b) Voltmeter
   (c) Ammeter (d) All of the above

4. What is the primary function of pliers?
   (a) Driving nails
   (b) Holding and gripping objects
   (c) Cutting metal
   (d) Measuring dimensions.

5. What is a Vernier Bevel Protractor used for?
   (a) Measuring the internal diameter of pipes
   (b) Measuring the flatness of surfaces
   (c) Measuring angles with high precision
   (d) Measuring the depth of grooves

6. Which of the following is the purpose of using an optical flat?
   (a) To check the flatness of surfaces
   (b) To measure the thickness of materials
   (c) To measure the dimensions of precision machined parts
   (d) To measure the angles of bevels

7. What is "bow" in the context of mechanical engineering?
   (a) A type of drill bit
   (b) A type of thread
   (c) The curvature or deviation from straightness in a component
   (d) The size of a bolt

8. What are Class I A and Class I B threads primarily used for?
   (a) Precision applications
   (b) General-purpose applications
   (c) High-stress applications
   (d) Low-torque applications

9. Which milling machine type is known for its versatility, as it can perform both horizontal and vertical milling operations?
   (a) Vertical milling machine
   (b) Horizontal milling machine
   (c) Universal milling machine
   (d) CNC milling machine

10. Which lathe part holds and rotates the workpiece being machined?

(a) Chuck (b) Tailstock

(c) Spindle (d) Toolpost

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain various safety precautions to be followed for Chemical Equipment.

Or

(b) List out the safety precautions while handling chemicals.

12. (a) Discuss about the Special Wrenches.

Or

(b) Explain various types of screw drivers.

13. (a) Write short notes on micrometer and its types.

Or

(b) With a neat sketch, explain the construction of Vemiar bevel protractor.

14. (a) Explain about the Tolerance Analysis.

Or

(b) Explain various classes of fits.

15. (a) Explain the construction of any one milling machine with a sketch.

Or

(b) List out the speed changing conditions for lathe.

Part C

$(5 \times 8 = 40)$

Answer **all** questions.

16. (a) List out the safety precautions when working with electricity in workshop.

Or

(b) What are the fire extinguishers used for fire caused by Solid, Liquid and metal?

17. (a) Discuss about the various types of hammers.

Or

(b) Explain standards of workmanship in workshop.

18. (a) Explain the construction and working of outside micrometer with a sketch.

Or

(b) List out the vernier caliper calibration procedure.

19. (a) Define following :

(i) Clearance (2)

(ii) Tolerance (2)

(iii) Allowance (2)

(iv) Limit (2)

Or

(b) Explain about the types of drills.

20. (a) Explain various operations performed in lathe.

Or

(b) Discuss about the types of gears.

C–1215

Sub. Code
11815

B.Sc. DEGREE EXAMINATION, NOVEMBER 2023.

First Semester

Aeronautical Science

AVIATION MATHEMATICS

(2023 onwards)

Duration : 3 Hours Maximum : 75 Marks

Part A $(10 \times 1 = 10)$

Answer **all** questions.

1. The minimum and maximum eigen values of the matrix $\begin{bmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$ are 1 and 4 respectively. What is the other eigen value?

   (a) 3 (b) 4

   (c) 0 (d) –2

2. If $B = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 4 \\ 0 & -2 & 5 \end{pmatrix}$. Then the product of the eigen values of B is ————.

   (a) 23 (b) 0

   (c) 15 (d) 1

3. Find the direction ratios of the line $2x = 3y = 5 - 4z$?

(a) $\langle 2, 3, 5 \rangle$ (b) $\langle 6, 4, -3 \rangle$

(c) $\langle 2, 3, -4 \rangle$ (d) None of these

4. What is the distance of the point $(1, 2, 3)$ form the plane $x - 3y + 2z + 13 = 0$?

(a) $\sqrt{14}$ (b) $\sqrt{13}$

(c) $\sqrt{11}$ (d) None of these

5. If $Y = X^x$, what is $\frac{dy}{dx}$ at $x = 1$ equal to?

(a) 0 (b) –1

(c) 2 (d) 1

6. The locus of center of curvature is called

(a) Involute

(b) Evolute

(c) Radius of curvature

(d) Envelope

7. If $y = \log \sin x$, then $\frac{dy}{dx}$ is _____

(a) $\log \cos x$ (b) $\frac{1}{\sin x}$

(c) $\tan x$ (d) $\frac{1}{\sin x} . \cos x$

8. If $u = \log_e\left(\frac{x^4 + y^4}{x + y}\right)$, the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is _____.

(a) 6 (b) 5

(c) 4 (d) 3

9. Which colour is used to illustrate actual progress in bar charts?

(a) red (b) black

(c) blue (d) green

10. Who was the first to use bar charts?

(a) Henry Williams (b) Henry Gantt

(c) Jane Gantt (d) Henry Joseph

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) If $\lambda_1, \lambda_2, \ldots\ldots, \lambda_n$ are eigen values of matrix A, then prove that the inverse A^{-1} has the eigen values $1/\lambda_1$, $1/\lambda_2, \ldots\ldots, 1/\lambda_n$.

Or

(b) Find the eigen values of the matrix $A = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 1 & 1 \\ -7 & 2 & -3 \end{pmatrix}$.

12. (a) Find the condition that the plane $lx + my + nz = p$ is a tangent plane to the conicoid $ax^2 + by^2 + cz^2 = 1$.

Or

(b) Find the equations to the line in which the plane $2x + y - z = 0$ cuts the cone $4x^2 - y^2 + 3z^2 = 0$.

13. (a) Find the radius of curvature at (x, y) for the curve $a^2 y = x^3 - a^3$.

Or

(b) Find the coordinate of the center of curvature on the parabola $y^2 = 4ax$ at any point (x, y).

14. (a) If u, v, w are functionally dependent functions of three independent variables x, y, z then prove that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$.

Or

(b) If $f(x, y) = 0$ then find $\frac{d^2 y}{dx^2}$.

15. (a) What is meant by program evaluation and review technique?

Or

(b) What is the difference between PERT and CPM.

Part C

$(5 \times 8 = 40)$

Answer **all** questions.

16. (a) Find the eigen values and eigen vectors of $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.

Or

(b) Diagonalise the matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & -2 & 1 \end{bmatrix}$ by means of orthogonal transformation.

17. (a) Find the shortest distance between lines l_1 and l_2 whose vector equation are $\vec{r} = \hat{i} + \hat{j} + \lambda\left(2\hat{i} - \hat{j} + \hat{k}\right)$ and $\vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu\left(3\hat{i} - 5\hat{j} + 2\hat{k}\right)$.

Or

(b) Find the equation of the two tangent planes to the sphere $x^2 + y^2 + z^2 = 9$ which pass through the line $x + y - 6 = 0 = x - 2z - 3$.

18. (a) Find the radius of curvature of the curve $r^2 \cos 2\theta = a^2$.

Or

(b) Find the envelope of the family of lines $x \cos^3 \alpha + y \sin^3 \alpha = a$ the parameter being α.

19. (a) If $u = f(r, s, t)$ and $r = x/y, s = y/z, t = z/x$ prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.

Or

(b) If $f(x, y) = \tan^{-1}(xy)$, compute an approximate value of $f(0.9, -1.2)$.

20. (a) Explain the advantages and disadvantages in CPM.

Or

(b) A project has the following times schedule :

Activity	Times in weeks	Activity	Times in weeks
(1-2)	4	(5-7)	8
(1-3)	1	(6-8)	1
(2-4)	1	(7-8)	2
(3-4)	1	(8-9)	1
(3-5)	6	(8-10)	8
(4-9)	5	(9-10)	7
(5-6)	4		

Construct the network and compute :

(i) T_E and T_L for each event.

(ii) Float for each activity

(iii) Critical path and its duration.